

Lecture 27: Oct 24

Last time

- Common Discrete Distribution

Today

- Common Discrete Distribution

Hypergeometric Distribution Suppose a population of N entities is made up of two types: M of the first type and $N - M$ of the second type. Suppose we take a sample of size K . We wish to know X , the number in the sample of the first type. The probability mass function of X is given by:

$$f_X(x) = \frac{\binom{M}{x} \binom{N-M}{K-x}}{\binom{N}{K}}$$

for $x = \max(0, M - N + K), \dots, \min(M, K)$.

The sample space is defined so that all binomial coefficients are valid. We must have:

$$0 \leq x \leq K, \quad 0 \leq x \leq M, \quad 0 \leq K - x \leq N - M$$

Often $K < M$ and $K < N - M$ so the range becomes $0 \leq x \leq K$.

Hypergeometric vs Binomial We can show that the limiting form of the hypergeometric pmf is the binomial pmf

$$\begin{aligned} \Pr(s) &= \frac{\binom{M}{s} \binom{N-M}{n-s}}{\binom{N}{n}} \\ &= \frac{\frac{M!}{s!(M-s)!} \frac{(N-M)!}{(n-s)!(N-M-n+s)!}}{\frac{N!}{n!(N-n)!}} \\ &= \frac{\frac{n!}{s!(n-s)!} \frac{M!}{(M-s)!} \frac{(N-M)!}{(N-M-n+s)!}}{\frac{N!}{(N-n)!}} \end{aligned}$$

Note

$$\begin{aligned} \frac{M!}{(M-s)!} &= \frac{M(M-1)(M-2) \dots (M-s)!}{(M-s)!} \\ &= M^s \left[1 \left(1 - \frac{1}{M}\right) \dots \left(1 - \frac{s-1}{M}\right) \right] \\ \frac{N!}{(N-n)!} &= N^n \left[1 \left(1 - \frac{1}{N}\right) \dots \left(1 - \frac{n-1}{N}\right) \right] \\ \frac{(N-M)!}{[(N-M)-(n-s)]!} &= (N-M)^{n-s} \left[1 \left(1 - \frac{1}{N-M}\right) \dots \left(1 - \frac{n-s-1}{N-M}\right) \right] \end{aligned}$$

Letting $N \rightarrow \infty, M \rightarrow \infty, \frac{M}{N} \rightarrow p$, we have

$$\begin{aligned}\Pr(s) &= \frac{\binom{M}{s} \binom{N-M}{n-s}}{\binom{N}{n}} \\ &\approx \binom{n}{s} \frac{M^s (N-M)^{n-s}}{N^n} \\ &= \binom{n}{s} \left(\frac{M}{N}\right)^s \left(1 - \frac{M}{N}\right)^{n-s} \\ &\rightarrow \binom{n}{s} p^s (1-p)^{n-s}\end{aligned}$$

In summary, we have

$$\begin{array}{lll}\text{Hypergeometric} & \rightarrow & \text{Binomial} \rightarrow \text{Poisson} \\ N \rightarrow \infty & & n \rightarrow \infty \quad \lambda = np \\ M \rightarrow \infty & & p \rightarrow 0 \\ \frac{M}{N} \rightarrow p & & np \rightarrow \lambda\end{array}$$

Geometric Distribution Consider a series of iid Bernoulli Trials with p = probability of success in each trial. Define a random variable X representing the number of trials until first success. Note X includes the trial at which the success occurs (one parameterization). Then, X has a geometric distribution.

- Sample space: $\{1, 2, \dots\}$
- pmf:

$$f(x) = \Pr(X = x) = \begin{cases} p(1-p)^{x-1} & x = 1, 2, \dots \\ 0 & \text{otherwise} \end{cases}$$

- cdf:

$$F(x) = \Pr(X \leq x) = \begin{cases} 0 & x < 1 \\ 1 - (1-p)^{\lfloor x \rfloor} & 1 \leq x \end{cases}$$

- Moments:

$$\begin{aligned}E(X) &= 1/p \\ \text{Var}(X) &= (1-p)/p^2\end{aligned}$$

Memoryless property. Suppose k and i are integers, and $k > i > 0$, then

$$\Pr(X > k | X > i) = \Pr(X > k - i)$$

Proof:

$$\begin{aligned}\Pr(X > k | X > i) &= \frac{\Pr(X > k)}{\Pr(X > i)} = \frac{(1-p)^k}{(1-p)^i} \\ &= (1-p)^{k-i} = \Pr(X > k - i)\end{aligned}$$

Example Suppose X is number of years you live, and X follows a geometric distribution, then

$$\begin{aligned}\Pr(\text{survive two more years}) &= \Pr(X > \text{current age} + 2 | X > \text{current age}) \\ &= \Pr(X > 2)\end{aligned}$$

This model is clearly too simple for human populations (since we do age).

Negative Binomial Distribution Still in the context of iid Bernoulli trials, define a random variable corresponding to the number of trials required to have s successes. We say $X \sim \text{Negbin}(s, p)$.

- Sample space: $\{s, (s + 1), \dots\}$
- pmf: for $x = s, s + 1, s + 2, \dots$

$$\begin{aligned}f(x) &= \binom{x-1}{s-1} p^{s-1} q^{x-s} \cdot p \\ &= \binom{x-1}{s-1} p^s q^{x-s}\end{aligned}$$

- cdf: no closed form
- Expectation: $EX = s/p$.
- Variance: $Var(X) = s(1-p)/p^2$

Notes

- Why the name? See Casella & Berger p.95.
- $X \sim \text{Negbin}(1, p)$ is the same as $X \sim \text{Geometric}(p)$
- $\text{Negbin}(n, p)$ is the same as the sum of n iid $\text{Geometric}(p)$ random variables

Other parameterizations The negative binomial distribution is sometimes defined in terms of the random variable Y = number of failures before the r th success. Then

- Sample space: $\{0, 1, 2, \dots\}$
- pmf

$$f(y) = \binom{r+y-1}{y} p^r q^y, \quad y = 0, 1, 2, \dots$$

- cdf: no closed form
- Expectation: $EY = r(1-p)/p$
- Variance: $Var(Y) = r(1-p)/p^2$

Negative binomial vs. Poisson The negative binomial distribution is often good for modeling count data as an alternative to the Poisson. In the previous parameterization, define

$$\lambda = \frac{r(1-p)}{p} \iff p = \frac{r}{r+\lambda}$$

Then we have

$$EX = \lambda$$

$$Var(X) = \frac{\lambda}{p} = \lambda(1 + \frac{\lambda}{r}) = \lambda + \frac{\lambda^2}{r}$$

For the Poisson we had that the variance equals the mean.

For the negative binomial, the variance is equal to the mean plus a quadratic term. Thus the negative binomial can capture overdispersion in count data.

In the previous parameterization, the pmf becomes

$$f(y) = \binom{r+y-1}{y} p^r q^y = \frac{(r+y-1)!}{y!(r-1)!} \left(\frac{r}{r+\lambda}\right)^r \left(\frac{\lambda}{r+\lambda}\right)^y$$

$$= \frac{\lambda^y r(r+1) \dots (r+y-1)}{y! (r+\lambda)^y} \left(1 + \frac{\lambda}{r}\right)^{-r}$$

Letting $r \rightarrow \infty$, we get

$$f(x) \rightarrow \frac{\lambda^x}{x!} e^{-\lambda}$$

So for large r , the negative binomial can be approximated by a Poisson with parameter $\lambda = r(1-p)/p$.

Common continuous distributions

Uniform Distribution A random variable X having a pdf

$$f(x) = \begin{cases} 1 & \text{for } 0 < x \leq 1 \\ 0 & \text{otherwise} \end{cases}$$

is said to have a *uniform distribution* over the interval $(0, 1)$.

The cdf is:

$$F(y) = \int_{-\infty}^y f(x)dx = \begin{cases} 0 & \text{for } y \leq 0 \\ y & \text{for } 0 \leq y \leq 1 \\ 1 & \text{for } y > 1 \end{cases}$$

- Unifrom; $Y \sim U[a, b]$
- sample space: $[a, b]$
- pdf:

$$f(y) = \begin{cases} \frac{1}{b-a} & \text{for } a < y \leq b \\ 0 & \text{otherwise} \end{cases}$$

- cdf:

$$F(y) = \int_{-\infty}^y f(x)dx = \begin{cases} 0 & \text{for } y \leq a \\ \frac{y-a}{b-a} & \text{for } a \leq y \leq b \\ 1 & \text{for } y > b \end{cases}$$

- moments:

$$E(Y) = (a + b)/2$$

$$Var(Y) = \frac{(b - a)^2}{12}$$

Notes

- The uniform extends to the continuous case the idea of equally likely outcomes.
- If $Y \sim U[0, 1]$, then $a + (b - a)Y \sim U[a, b]$

Exponential Distribution Denoted $X \sim Exp(\lambda)$:

- sample space: $x \geq 0$
- pdf:

$$f(x) = \begin{cases} \lambda e^{-\lambda x} & \text{for } x \geq 0 \\ 0 & \text{otherwise} \end{cases}$$

- cdf:

$$F(x) = \int_{-\infty}^x f(y)dy = \begin{cases} 1 - e^{-\lambda x} & \text{for } x \geq 0 \\ 0 & \text{for } x < 0 \end{cases}$$

- moments:

$$E(X) = 1/\lambda$$

$$Var(X) = 1/\lambda^2$$

$$M_X(t) = \lambda/(\lambda - t), \quad t < \lambda$$

Interpretation The exponential can be derived as the waiting time between Poisson events. Suppose that the number of events in a unit interval of time follows a Poisson(λ) distribution. Then, let Y be the time until the first event.

$$\Pr(Y > t) = \Pr(0 \text{ events in } [0, t])$$

and the number of events in $[0, t]$ follows a Poisson distribution with parameter λt . Therefore,

$$\Pr(Y > t) = e^{-\lambda t}.$$

The cdf of Y is

$$F(t) = 1 - \Pr(Y > t) = 1 - e^{-\lambda t}$$

and hence the density is $f(t) = \lambda e^{-\lambda t}$.

Alternative parameterization Many books write the density as

$$f(y) = \begin{cases} \frac{1}{\theta} e^{-y/\theta} & \text{for } y \geq 0 \\ 0 & \text{otherwise} \end{cases}$$

so that $E(Y) = \theta$ and $Var(Y) = \theta^2$. In this case $\theta = 1/\lambda$ is called the *mean parameter*, while $\lambda = 1/\theta$ is called the *rate parameter*.

Memoryless property The exponential has a memoryless property, just like the geometric.

$$\Pr(Y > s + t | Y > t) = \Pr(Y > s)$$

Same interpretation as the geometric for continuous time:

- The probability of an event in a time interval depends only on the length of the interval, not the absolute time of the interval.
- The underlying Poisson process is stationary: the rate λ is constant. (In the geometric case, the probability, p of getting an event in every discrete time unit is constant).

Shifted exponential Let $X \sim Exp(\lambda)$ and $Y = X + v, v \in \mathbb{R}$. Then, Y has the *shifted exponential distribution* with pdf:

$$f(y) = \begin{cases} \lambda e^{-(y-v)\lambda} & \text{for } y \geq v \\ 0 & \text{otherwise} \end{cases}$$

Interpretation:

- $v > 0$: Event is delayed
- $v < 0$: The news of the event is delayed

Does the shifted exponential maintain the memoryless property?

Double exponential The *double exponential distribution* is formed by reflecting an exponential distribution around zero. It has pdf:

$$f(x) = \frac{1}{2} \lambda e^{-\lambda|x|}, \quad x \in \mathbb{R}$$

Laplace distribution Suppose X has the above distribution with $\lambda = 1$. Now let $Y = \sigma X + \mu, \mu \in \mathbb{R}$ (shifting) and $\sigma > 0$ (scaling). Then Y has the *Laplace distribution* with pdf:

$$f_Y(y) = \frac{1}{2\sigma} \exp\left(-\frac{|y - \mu|}{\sigma}\right)$$

with moments

$$EY = \mu, \quad Var(Y) = 2\sigma^2$$

The Laplace distribution provides an alternative to the normal for centered data with fatter tails but all finite moments.

Normal Distribution Introduced by De Moivre (1667 - 1754) in 1733 as an approximation to the binomial. Later studied by Laplace and others as part of the Central Limit Theorem. Gauss derived the normal as a suitable distribution for outcomes that could be thought of as sums of many small deviations.

- Sample space: $\mathbb{R} = (-\infty, \infty)$
- pdf: For $Y \sim N(\mu, \sigma^2)$,

$$f(y) = \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{(y-\mu)^2}{2\sigma^2}} \quad -\infty < y < \infty$$

- cdf: There is no closed form.
- When $\mu = 0$ and $\sigma = 1$, the distribution is called *standard normal*:

$$\Phi(y) = \Pr(Y \leq y), \quad \Phi(-y) = 1 - \Phi(y)$$

- Mean:

$$EY = \mu$$

- Variance:

$$Var(Y) = E(Y - \mu)^2 = \sigma^2$$

- Higher central moments:

$$E(Y - \mu)^m = \begin{cases} \frac{m!}{2^{m/2}(m/2)!} \sigma^m & m \text{ is even} \\ 0 & m \text{ is odd} \end{cases}$$

- In particular:

$$\begin{aligned} \mu_3 &= E(Y - \mu)^3 = 0 \text{ (Skewness)} \\ \mu_4 &= E(Y - \mu)^4 = 3\sigma^4 \end{aligned}$$

- Moment generating function:

$$M_Y(t) = \exp(\mu t + \sigma^2 t^2 / 2)$$

Standardization

$$Y \sim N(\mu, \sigma^2) \iff Z = \frac{Y - \mu}{\sigma} \sim N(0, 1)$$

Shifting and scaling:

$$Z \sim N(0, 1) \iff Y = \sigma Z + \mu \sim N(\mu, \sigma^2)$$

Notes

- Normal distribution is useful in many practical settings. For example, measurement error.

- Plays an important role in *sampling distributions* in *large samples*, since the Central Limit Theorem says that the sums of independent identically distributed random variables are approximately normal
- There are many important distributions that can be derived from functions of normal random variables (e.g., χ^2 , t , F). We will briefly present the pdf's and sample spaces of these distributions.

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χ^2 distribution If $Z \sim N(0, 1)$, then $X = Z^2$ has the χ^2 distribution with 1 degree of freedom. More generally, we have the χ^2 distribution with v degrees of freedom with pdf:

$$f(x) = \frac{(x/2)^{\frac{v}{2}-1} e^{-x/2}}{2\Gamma(v/2)}, \quad x > 0$$

where $\Gamma(a)$ is the complete gamma function,

$$\Gamma(a) = \int_0^{\infty} x^{a-1} e^{-x} dx$$

The $\chi^2(v)$ distribution is a special case of the gamma distribution, so it is easier to derive its properties from the gamma.

Facts about the Gamma function

- $\Gamma(a+1) = a\Gamma(a), a > 0$
- $\Gamma(1) = 1$
- $\Gamma(n) = (n-1)!$
- $\Gamma(1/2) = \sqrt{\pi}$